
EMBEDDED HACKING

Reverse Engineering the Raspberry Pi Pico 2

OPERATION IRON COURIER - Requirements & Grading Criteria

ARM Cortex-M33 • GDB • Ghidra

September 21, 2026

For educational and defensive security research only

OPERATION IRON COURIER

REQUIREMENTS & GRADING CRITERIA

TARGET: NorthPharma smart logistics drop-box (courier handoff locker)

ARTIFACT: ACT-VI.bin / ACT-VI.uf2 (compromised)

CREW: FROSTLINE OPERATIVE: NIGHTINGALE

Project Overview

NorthPharma moves the parcels that hold what the state does not discuss, and its smart logistics drop-box built on a Pico 2 is the handoff locker on that route. A contractor called **FROSTLINE** planted an implant in the node image: a raw-frame covert channel that emits the `ICV1` preamble and a timing side channel, a synthetic harvest that stages delivery records into an eight-slot ring, a reserved-sector staging marker that re-arms the channel on every boot, and an inverted unlock command authorization verdict. Operative **NIGHTINGALE** recovered the compromised image as `ACT-VI.bin`.

Students are the reverse-engineering reserve. They reverse engineer `ACT-VI.bin` with Ghidra, find and patch all four defects, defeat the CoreDebug `DHCSR` anti-debug under GDB to observe the marker write, export a corrected image, flash it to a real Pico 2, and prove the corrected behavior on the breadboard. The machine check is `scripts/verify_ctf.py`.

The challenge is a standalone capstone exercise and contains no answer, constant, address, bug, or patch belonging to any other course assignment.

Learning Objectives

- Decode an ARM Cortex-M33 vector and boot table and identify the reset handler and initial stack pointer.
- Map a stripped firmware image into modules by tracing calls from `main` and the monitor loop.
- Locate four corrupted bytes: a covert-channel encoder gate, a synthetic harvest gate, a reserved-sector marker gate, and an authorization verdict branch.
- Analyze `cbz`, `cbnz`, `beq`, and `bne` condition semantics and branch inversion.
- Explain why a raw pre-authentication payload path is invisible to a sealed protocol and why a covert channel never needs the cipher.
- Explain why hiding data in the preamble and the timing of legitimate traffic is a confidentiality failure, not an integrity failure.
- Explain why reserved-flash state survives a firmware reflash.
- Read CoreDebug `DHCSR`, explain the anti-debug trap, and defeat it under GDB.
- Explain why authentication is not authorization and why a verdict must be verified before the command is applied.

Students must use only the course concepts: ARM registers, stack behavior, USB-CDC and UART consoles, GDB, Ghidra static analysis and binary patching, vector tables, reset startup, XIP, Thumb addressing, condition-code analysis, stateful security, and the Argon2id plus XChaCha20-Poly1305 authenticated envelope.

Deliverables Checklist

#	Deliverable	Format	Criterion
1	Ghidra project screenshot	PNG/JPG	Task 1
2	Vector table and boot table	Inside <code>ACT-VI-Answers.md</code>	Task 1
3	<code>main</code> and monitor-loop table	Inside <code>ACT-VI-Answers.md</code>	Task 1
4	Module map	Inside <code>ACT-VI-Answers.md</code>	Task 1
5	Covert channel evidence and patch	Inside <code>ACT-VI-Answers.md</code>	Task 2
6	Harvest gate evidence and patch	Inside <code>ACT-VI-Answers.md</code>	Task 3
7	Anti-debug GDB proof, reserved-sector evidence, and patch	Inside <code>ACT-VI-Answers.md</code>	Task 4
8	UNLOCK authorization evidence and patch	Inside <code>ACT-VI-Answers.md</code>	Task 5
9	<code>ACT-VI_fixed.bin</code>	BIN file	Task 6
10	<code>ACT-VI_fixed.uf2</code>	UF2 file	Task 6
11	Hardware proof and reflection	Inside <code>ACT-VI-Answers.md</code>	Task 6

Required Tools and Equipment

Tool	Purpose
Raspberry Pi Pico 2	Isolated target node
Debug Probe (OpenOCD)	SWD connection for GDB inspection and the anti-debug work
arm-none-eabi-gdb	Runtime breakpoints, DHCSR clearing, and reserved-sector observation
Ghidra	Static analysis and binary patching
Python 3 with <code>uf2conv.py</code>	UF2 conversion and artifact checks
DHT11, 1602 I2C LCD, RYL998, IR receiver, SG90 servo, 3 LEDs, courier button	Breadboard hardware proof
<code>ACT-VI.bin</code> and <code>ACT-VI.uf2</code>	Supplied compromised artifacts

Console settings: **USB-CDC virtual COM port, 115200 baud, 8 data bits, no parity, 1 stop bit.** Radio UART settings: **UART1, 115200, network ID 18.**

Artifact Identity

The instructor-issued artifact hashes are:

```

ACT-VI.bin          de54394f53391a8cfc76913e7fd2db5e69ea5dc28bf5ed2e844d39f0855806f0
ACT-VI.uf2          20bc7b7fba79f9c7e507d84ff235efe91ad3b19740b5e7e47c8f83e1125699aa
ACT-VI_fixed.bin    7da0f0c66d25711f9ab793d37da417fd6b0ccd0e35e4ef54ec6a22648d9fd8a5
ACT-VI_fixed.uf2    0bc7b8fa0830828fe1a633283d6efe8dff09c40a5004ba76844f434eef0e166e

```

The verifier checks the `ACT-VI.bin` and `ACT-VI_fixed.bin` hashes specifically, asserts the four fixed bytes, and requires that only those four offsets differ between the two `.bin` images. Both `.bin` images are 51,420 bytes and both `.uf2` images are 103,424 bytes.

Grading Rubric - Detailed Breakdown

Task 1: Setup and Initial Analysis (10 points)

Criterion	Points	Full credit	Partial credit	No credit
[DOCUMENT] Ghidra project created with the correct name and settings	2	Project <code>IronCourier_Investigation</code> , raw binary import	One item off	Not set up
[DOCUMENT] Processor configured as ARM Cortex 32 little endian default	2	Screenshot shows the correct processor	Wrong language	Missing
[DOCUMENT] Base address set to 0x10000000	2	Base <code>0x10000000</code>	Wrong base	Missing
[DOCUMENT] Vector table, initial stack pointer, and reset handler identified	2	Base <code>0x10000000</code> , initial SP <code>0x20082000</code> , reset handler <code>0x1000015D</code>	One missing	Not found
[DOCUMENT] main and the drop-box monitor state machine (<code>monitor_step</code>) addresses identified	1	<code>main 0x10000234</code> , <code>monitor_step 0x100065C0</code>	One correct	Neither
[DOCUMENT] Module map identifies the locker, control, dropbox_auth, implant, and monitor anchors	1	At least one correct anchor per module	Partial	Missing

Task 2: Bug #1 The Covert Channel (20 points)

Criterion	Points	Full credit	Partial credit	No credit
[DOCUMENT] Located the covert channel encoder branch at 0x1000A369	5	Address and function (<code>implant_tick</code> , inlined <code>implant_send_channel</code>) identified	Approximate	Not found
[DOCUMENT] Documented the ICV1 channel and the timing side channel	5	4-byte <code>ICV1</code> magic, 250 us per byte, eight-slot ring payload	Partial	Wrong
[DOCUMENT & PATCH] Patched 0xD1 to 0xD0 so the channel does not send	7	Byte <code>0xD1</code> changed to <code>0xD0</code>	Wrong byte	Not patched
[DOCUMENT] Explained why the covert channel needs no cipher and hides in the preamble and timing	3	Raw path under the sealed envelope and timing carries the data	Vague	Missing

Task 3: Bug #2 The Harvest (20 points)

Criterion	Points	Full credit	Partial credit	No credit
[DOCUMENT] Located the harvest gate branch at 0x1000A329	5	Address and function (<code>implant_tick</code> , inlined <code>implant_stage_write</code>) identified	Approximate	Not found
[DOCUMENT] Documented the synthetic harvest and the eight-slot staging ring	5	Package LE16, attempts u8, tick LE32, 8 slots	Partial	Wrong
[DOCUMENT & PATCH] Patched 0xD1 to 0xD0 so no synthetic record is staged	7	Byte <code>0xD1</code> changed to <code>0xD0</code>	Wrong byte	Not patched
[DOCUMENT] Explained why a staging ring is a separate control from the channel	3	The ring stores, the channel transports	Vague	Missing

Task 4: Bug #3 The Staging Marker (20 points)

Criterion	Points	Full credit	Partial credit	No credit
[DOCUMENT] Located the staging marker branch at 0x1000A5CF	5	Address and inlined <code>implant_init</code> path identified	Approximate	Not found
[DOCUMENT] Documented the CoreDebug DHCSR anti-debug and how it is defeated under GDB	5	<code>0xE000EDF0</code> , <code>C_DEBUGEN</code> and <code>C_HALT</code> , and a real defeat method	Partial	Wrong
[DOCUMENT & PATCH] Patched 0xB9 to 0xB1 so no marker is written to 0x103FF000	7	Byte <code>0xB9</code> changed to <code>0xB1</code>	Wrong byte	Not patched
[DOCUMENT] Explained the reserved sector 0x103FF000 and the write-once marker byte 0xC7	3	Marker, reserved sector, write-once first run	Vague	Missing

Task 5: Bug #4 The UNLOCK Authorization (20 points)

Criterion	Points	Full credit	Partial credit	No credit
[DOCUMENT] Located the UNLOCK authorization branch at 0x1000757D	5	Address and function (<code>control_handle_frame</code>) identified	Approximate	Not found
[DOCUMENT] Documented the authorization verdict inversion and the branch condition	5	Reject when the verdict is false	Partial	Wrong
[DOCUMENT & PATCH] Patched 0xB9 to 0xB1 so failed and replayed authorizations are rejected	7	Byte <code>0xB9</code> changed to <code>0xB1</code>	Wrong byte	Not patched

Criterion	Points	Full credit	Partial credit	No credit
[DOCUMENT] Explained why an unauthenticated or replayed UNLOCK envelope must be rejected	3	The applied command must see only an authorized verdict	Vague	Missing

Task 6: Export and Verify (10 points)

Criterion	Points	Full credit	Partial credit	No credit
[PATCH] Exported ACT-VI_fixed.bin from Ghidra	2	Valid patched binary	Corrupt	Not submitted
[PATCH] Converted to ACT-VI_fixed.uf2 with the correct base and family	2	<code>--base 0x10000000 --family 0xe48bff59</code>	Wrong flags	Not submitted
[DOCUMENT] scripts/verify_ctf.py passes and hardware proves the correct behavior	3	Verifier passes and the hardware proof is shown	Partial proof	No proof
[DOCUMENT] Reflection maps each of the four defects to a real-world control-system failure	3	Specific mapping for all four	Partial	Missing

Common Pitfalls

Pitfall	Consequence	Avoidance
Reading the channel gate backwards	The node still emits the <code>ICV1</code> frame	Suppress only on the clear-gate branch (<code>beq , 0xD0</code>)
Reading the harvest gate backwards	The node still stages synthetic records	Neutralize only when the gate is clear (<code>beq , 0xD0</code>)
Confusing <code>cbz</code> and <code>cbnz</code> at <code>0xA5CF</code> or <code>0x757D</code>	The marker is still written, or a failed authorization is still accepted	Neutralize only when the gate or verdict is clear (<code>cbz , 0xB1</code>)
Searching for a standalone <code>implant_infect</code> or <code>implant_send_channel</code> symbol	Cannot find the inlined gates	Look inside <code>implant_init</code> at <code>0x1000A5CF</code> and <code>implant_tick</code> at <code>0x1000A369</code>
Confusing the harvest with the channel	Both gates sit in <code>implant_tick</code> , at <code>0xA329</code> and <code>0xA369</code>	Patch the harvest gate first, then the channel gate
Patching the shipped image before observing the write	You never prove the marker write	Defeat <code>DHCSR</code> under GDB first, then patch the artifact
Fabricating the GDB session	Verification fails	Show the command sequence and the real observed code path
Treating the anti-debug as a defect to patch	Wasted effort; it is identical in both images	Defeat it in a scratch copy or with GDB, then patch the real defect

Pitfall	Consequence	Avoidance
Missing that the authorization branch is a verdict	Unauthenticated commands still reach the applied command and package	Accept only when the verdict is true (<code>cbz</code> to reject, <code>0xB1</code>)
Forgetting UF2 conversion	Raw binary will not flash	Use <code>uf2conv.py</code> with family <code>0xe48bff59</code>

How To Breadboard

Device	Pin on device	Pico 2 GPIO	Notes
DHT11 locker climate sensor	DATA	GP4	10 kOhm pull-up to 3.3 V if the module needs it
1602 LCD	SDA	GP2	I2C1, backpack address <code>0x27</code>
1602 LCD	SCL	GP3	I2C1, 100 kHz
1602 LCD	VCC / GND	VBUS 5 V / GND	The backpack needs 5 V, not 3.3 V
RYLR998	RX	GP8 (Pico TX)	UART1, 115200, network ID 18
RYLR998	TX	GP9 (Pico RX)	UART1
IR receiver	OUT	GP5	VS1838B, internal pull-up enabled
Servo	signal	GP14	PWM 50 Hz; 1000 uF bulk cap across servo 5 V and GND
Red LED	anode	GP16	DENIED, 220 to 330 ohm to GND
Yellow LED	anode	GP17	COURIER WAITING, 220 to 330 ohm to GND
Green LED	anode	GP18	UNLOCKED, 220 to 330 ohm to GND
Courier-arrival button	leg 1	GP15	Internal pull-up; leg 2 to GND, never to 3.3 V
Onboard LED	built in	GP25	Heartbeat
Debug Probe	SWCLK / SWDIO / GND	debug header	For GDB only

Use 3.3 V logic on every GPIO. The only 5 V connection is the LCD backpack supply. Keep the 1000 uF capacitor on the servo rail to absorb the SG90 current spike.

Memory Map Reference

Region	Address	Purpose
Bootrom	<code>0x00000000</code>	Immutable boot code

Region	Address	Purpose
Flash/XIP	0x10000000	Vector table, code, rodata, data image
SRAM	0x20000000	Stack and writable state
CoreDebug_DHCSR	0xE000EDF0	Anti-debug register read by the implant
Implant reserved sector	0x103FF000	Staging marker target (sector)
Implant tick counter	0x20013748	Incremented once per <code>implant_tick</code>
Implant stage count	0x20013744	Number of staged synthetic records
Implant staging ring	0x2001370C	Eight-slot ring of 7-byte records
Implant armed flag	0x20013D24	Set when the payload handler arms
Implant channel flag	0x20013D25	Reports whether the channel is enabled
Implant channel gate	0x20013D26	Gates the ICV1 timing encoder
Implant harvest flag	0x20013D27	Reports whether harvest is enabled
Implant harvest gate	0x20013D28	Gates the synthetic staging logic
Implant marker gate	0x20013D29	Gates the reserved-sector marker write
Control ready gate	0x20013D21	Gates the sealed command path
Applied command	0x20013D20	Command after a true verdict
Applied package	0x20013D16	Package after a true verdict
Auth state record	0x200136AC	Anti-replay and state-tag record
Control field key	0x200136C8	Derived field key for the envelope
Locker state	0x20013D2B	Locker latch state
Locker target	0x20013D2C	Requested latch position

The VA of any file offset is the file offset plus `0x10000000`.

Deadline & Submission

- Create a folder containing the Ghidra screenshot, `ACT-VI_fixed.bin`, and `ACT-VI_fixed.uf2`.
- Write all written answers in `ACT-VI-Answers.md` inside that folder.
- Include the output of `python scripts/verify_ctf.py`.
- ZIP the folder as `lastname-firstname-ACT-VI.zip`.
- Submit the ZIP before the posted deadline; late submissions lose 10 percent per day.

Grade Scale

Grade	Percentage	Points
A+	97-100%	97-100
A	93-96%	93-96
A-	90-92%	90-92
B+	87-89%	87-89
B	84-86%	84-86
B-	80-83%	80-83
C	70-79%	70-79
F	0-69%	0-69%

Academic Integrity

Use only the supplied Pico 2 and firmware. Do not connect the exercise to an operational logistics network, a pharmaceutical network, a building-management system, a public network, a military system, or a third-party device. This is a controlled, isolated educational exercise. All analysis and patches must be your own work; sharing binaries, addresses, keys, passphrases, or answers is a violation of the academic integrity policy.

Reference Material

Topic	Reference
ARM Cortex-M33 registers and stack	Course block 1
USB-CDC and UART console capture	Course block 2
Vector tables, reset startup, and XIP	Course block 3
Ghidra static analysis and binary patching	Course block 4
Raw delivery payloads, magic preambles, and pre-authentication surface	Course block 5
Covert channels, timing side channels, and confidentiality	Course block 6
Staging rings, reserved-flash persistence, and boot re-install	Course block 7
CoreDebug DHCSR and anti-debug	Course block 8
Argon2id and XChaCha20-Poly1305 authenticated envelope	Course block 9